

Research paper

A data-driven machine learning framework for predicting total agricultural food grain yield

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ABSTRACT

India's agricultural sector is experiencing significant shifts due to climate variability, evolving farming practices, and changing crop patterns. While previous studies have explored crop yield prediction using machine learning commonly adopt image-based Machine learning techniques, many relied on manual hyperparameter tuning and rarely applied these techniques to decision tree based algorithms, limiting reproducibility and optimization. To address this gap, our study presents a consistent, data-driven approach using automated hyperparameter configuration and a moderate-scale, publicly available dataset from the Reserve Bank of India (1967–2024), covering key crops such as Groundnut, Rapeseed, Mustard, Soybean, Coffee, Sugarcane, and Tea. Four ensemble models - Gradient Boost, XGBoost, LightGBM, and CatBoost are used to predict total food grain yield. The model pipeline incorporates five-fold cross-validation, GridSearchCV for hyperparameter tuning, early stopping to prevent overfitting, and SHapley Additive exPlanations(SHAP) for interpretability. Model performance is evaluated using Mean Error (ME), Mean Absolute Error (MAE), Mean Percentage Error (MPE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and Coefficient of Determination (R^2). Gradient Boost achieved the highest accuracy, followed by CatBoost, XGBoost, and LightGBM. The results demonstrate the reliability and scalability of ensemble models and provide a strong foundation for future integration of geographic and climatic variables into yield forecasting systems.

1. Introduction

The Economic Survey 2023-24 highlights that, at current prices, the agriculture sector contributes 17.7% to the total Gross Value Added (GVA) in FY24. Despite a slower growth rate, the sector's GVA continues to increase, even under the pressure of irregular weather patterns and uneven monsoon distribution across regions in 2023, which negatively impacted overall production. The World Health Organization (WHO) emphasizes the urgent need for stronger and more systematic actions to increase food grain production, reform agricultural policies, and invest in global nutrition. As global demand for agricultural products continues to rise, it becomes increasingly important to develop farmland effectively and boost overall food grain yields.

The importance of agriculture in assuring global food security and economic development is underscored by the increasing effects of climate change [1]. The prediction of crop yields has been substantially improved by the incorporation of data-driven techniques, such as machine learning (ML). A recent review has emphasized the widespread

adoption of ML methods, particularly regression, tree-based, and neural network models, for the prediction of palm oil yield and other crop outcomes. Key features of these models include temperature, humidity, and nutrient levels [2]. At the macro level, long-term agricultural productivity in Europe identifies inclusive institutions, innovation-friendly agricultural policies, skilled human capital, and trade openness as key long-term drivers of productivity growth between 1950 and 2005 [3]. In another context, research undertaken in Tanzania demonstrates that receiving agricultural extension advice significantly enhances productivity and supports the adoption of improved farming techniques [4]. Further, a study in India explores the application of ML models such as Gradient Boost, Random Forest, SVM, and Decision Tree for predicting the yields of crops like maize, potatoes, rice, and wheat using features including weather data, pesticide usage, and historical yields [5]. Similarly, [6] A study developed a crop yield prediction model for Tamil Nadu using statistical techniques, confirming the effectiveness of simple regression-based models with weather and soil input features. Advanced models such as XGBoost achieved the highest R^2 value of

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0.97 for predicting yields of tropical fruits like bananas and arecanut [7]. In addition, ensemble frameworks with entropy-based uncertainty quantification achieved 99.54% accuracy across 22 crop classes, boosting the reliability of crop recommendations [8].

A survey on deep learning models in agriculture reveals that these approaches outperform traditional image processing methods in tasks such as crop disease detection and land-use classification. However, their application often demands large-scale labeled datasets and intensive computational resources [9]. Climate-related agricultural studies offer further insights. For instance, a study conducted in Benin shows that temperature fluctuations negatively affect economic growth, while food production continues to positively correlate with agricultural development [10]. In Pakistan, researchers found that cereal cropland area and rainfall had a favorable impact on productivity, whereas rising temperatures showed a long-term detrimental effect [11]. A case study in Kerala, India, highlights that crop yields have increased despite no significant trend in rainfall, underlining the importance of irrigation and the need for sustainable water management strategies [12].

Beyond agriculture, ML models have been successfully applied in materials science and environmental engineering. For instance, biochar yield has been predicted using ML models trained on pyrolysis variables such as temperature and ash content [13]. In the construction industry, ML techniques have been used to predict the compressive strength of concrete incorporating Palm Oil Fuel Ash (POFA). A hybrid XGB-LGBM model achieved a prediction accuracy with $R^2 = 0.976$ [14]. Hybrid ML models also show promise across other scientific disciplines. For example, the GWO-XGB model has been employed for ultra-high-performance concrete (UHPC) design [15], HCCIA-KELM for predicting concrete strength [16], and RAGN-R for advanced materials [17]. An optimization framework like AL-Stacked ML has also demonstrated 98.9% accuracy in supporting eco-friendly UHPC development [18].

Based on these insights, the present study sets out to:

- Employ ensemble machine learning models including Gradient Boost, XGBoost, CatBoost, and LightGBM to predict total agricultural food grain yield using historical crop production data.
- Utilize precise preprocessing and validation methodologies, including GridSearchCV, Early Stopping, and 5-fold cross-validation to minimize overfitting and improve the robustness of the model.
- Evaluate Model performance using various metrics: ME(Mean Error), MAE (Mean Absolute Error), MPE (Mean Percentage Error), RMSE(Root Mean Square Error), MSE(Mean Square Error), and R^2 .
- Integrate SHAP to interpret model predictions and highlight the most influential features driving yield outcomes.

This research is organized as follows: related work of ensemble ML is discussed in section 1.1, and the materials and methods discussed in section 2, section 3 describes results and discussion, section 4 conclusion.

1.1. Related works

Effective agricultural planning, sustainable resource use, and national food security depend critically on accurate prediction of overall agricultural food grain production. Since food grains are a basic part of socio-economic stability and a staple component of human nutrition, reliable prediction models are crucial for guiding agricultural practices and reducing any hazards connected to food shortages. Key players like farmers, legislators, and agribusiness companies may use yield prediction to make data-driven decisions improving output, lowering waste, and supporting food system resilience. Recent advances in modeling and prediction of total food grain output are presented in this part. Aiming at enhancing prediction accuracy and broadening existing knowledge of food grain yield dynamics, it synthesizes a wide spectrum of methods, data sets, and technological breakthroughs. The literature emphasizes increasingly the dynamic and multidisciplinary character of modern

agricultural yield forecasting research by reflecting a rising attention on the integration of data science and machine learning approaches [19].

A study on Europe (1950–2005) employs econometric modeling to calculate Total Factor Productivity (TFP) and identifies inclusive institutions, innovation-friendly policies, skilled human capital, and trade openness as key contributors to sustained productivity growth [3]. In Tanzania, analysis of 2019/2020 agricultural data using advanced econometric methods instrumental variables, extended regression, and endogenous switching regression demonstrates that access to extension services significantly improves productivity and food security for small-holder farmers by promoting better farming practices [4]. In Benin, a long-term study (1961–2021) using ARDL and Granger causality methods reveals that agricultural value-added and food production enhance economic growth, while temperature variations hinder it; additionally, livestock production shows bidirectional causality with both agriculture and economic growth [10]. In Wayanad, Kerala, despite no significant rainfall trends between 1989 and 2019, improved crop yields result from effective irrigation, emphasizing the need for climate-resilient agriculture and sustainable water management [12]. Similarly, a time series analysis in Pakistan (1961–2018) using the ARDL model finds that cereal cropland and rainfall positively influence productivity, while rising temperatures in the long run negatively impact it. The study underscores the importance of climate adaptation, optimized land use, and sustainable agricultural practices [11].

A data-driven approach predicted smartphone prices using 21 key features and multiple machine learning (ML) algorithms, with Support Vector Machine (SVM) achieving the highest accuracy of 97%, followed by K-Nearest Neighbor (KNN) at 94% [20,21]. To predict the mechanical properties of advanced materials like PAC and FRC, researchers developed RAGN-R, a surrogate ensemble model combining Random Forest, AdaBoost, and Gradient Boosting with Ridge regression stacking. It outperformed six standard models, reaching 97.7% accuracy [17]. In a comparison of eight species distribution models using 2,729 Swiss freshwater samples, complex ML models slightly outperformed statistical models during cross-validation but exhibited overfitting, reducing their generalizability and ecological reliability [22]. A study on breast cancer classification using clinical and genomic data showed that ensemble models like AdaBoost, GBM, and RGF outperformed individual models, with LGBM achieving an F1 score of 99.2% and accuracy of 98.9% [23]. In optimizing friction stir welding (FSW) parameters for AA2050-T8, Random Forest Regression achieved the highest predictive accuracy with an R^2 of 0.9266, validated using K-Fold cross-validation [24]. Another study developed a hybrid XGBoost–Grey Wolf Optimizer (GWO) model to predict ultra-high-performance concrete (UHPC) properties, achieving R^2 values of 98.4% for nanomaterials and 94.8% for micromaterials, while significantly reducing prediction errors [25]. A hybrid MLP–Q-learning model predicted the Remaining Useful Life (RUL) of aircraft engines more accurately than standalone models, enabling early failure predictions with only 2% late results and improving accuracy by 15% [26]. To predict biochar yield, researchers used ML models including Random Forest, AdaBoost, Decision Tree, and KNN; Decision Tree performed best with an R^2 of 0.771, and SHAP analysis identified residence time, pyrolysis temperature, and ash content as key features [13]. An optimized Kernel Extreme Learning Machine (KELM) model, enhanced with the Hybrid Chaotic Crested Ibis Algorithm (HCCIA), accurately predicted concrete strength by avoiding local optima and overfitting, offering a scalable solution [16]. For Parkinson's disease prediction, an SVM model combined with PCA, SMOTE, and GridSearchCV achieved 97.44% accuracy and 100% sensitivity on a small OpenML dataset [27], while a G-LSTM model for diabetes diagnosis attained 97.12% accuracy using SMOTE-enhanced LSTM with GridSearchCV optimization [28]. In financial modeling, a study evaluated k-fold cross-validation for selecting ML models predicting corporate bankruptcy and found that 67% of performance variability stemmed from data split choices, advising caution in interpretation [29]. In heart disease detection, researchers applied k-fold cross-validation and SMOTE to enhance model performance,

achieving 95.47% accuracy using an AdaBoost–Random Forest ensemble [30]. A CAD diagnosis system using KNN and ANN achieved 96.78% average accuracy, with high AUC and F1-scores, validated on Cleveland and Z-Alizadeh Sani datasets [31]. In analyzing different k values for cross-validation, $k = 7$ provided slightly better validation accuracy and AUC with lower computational cost compared to $k = 10$ [32]. For flood routing prediction in the Büyük Menderes River, the PSO-ANN hybrid model outperformed LSTM and other alternatives, achieving the best accuracy (e.g., $NSE = 0.941$) [33]. In predicting the compressive strength and tensile strain of engineered cementitious composites (ECC), CatBoost and XGBoost reached R^2 values up to 0.96. SHAP analysis highlighted the water-to-binder ratio and fiber elongation as the most influential variables [34]. A sensorless method using GRU-based recurrent neural networks accurately estimated projectile positions in electromagnetic launchers, achieving 94.88% accuracy and demonstrating feasibility in sensor-limited environments [15]. A hybrid XGB-LGBM model predicted concrete strength with Palm Oil Fuel Ash (POFA) with the highest accuracy ($R^2 = 0.976$), identifying water-to-binder ratio as the dominant factor [14]. For eco-friendly alkali-activated UHPC (AA-UHPC), researchers proposed an active learning-based stacked ML model (AL-Stacked ML-3) that achieved 98.9% accuracy and improved predictive performance by 4.1%. A GUI was also developed for validation and application [18]. Finally, a comparative study of cross-validation techniques showed that while k -fold was most efficient, Leave-One-Out and Repeated k -folds offered higher accuracy and sensitivity for tuned models, such as SVM and Bagging, depending on dataset characteristics [35]. A review on palm oil yield prediction highlighted the application of machine learning (ML) algorithms, discussing global trends, key features, and forecasting models. It also outlined future directions like remote sensing and proposed an efficient ML-based architecture for palm oil yield prediction [2]. A survey of 40 studies examined deep learning applications in agriculture and food production, showing that deep learning consistently outperformed traditional techniques in accuracy and reliability [9]. Another study applied ML models Gradient Boost, Random Forest, SVM, and Decision Tree to predict yields of major Indian crops using weather, pesticide, and historical data, enabling better input optimization [5]. To enhance crop recommendation, an ensemble ML framework with entropy-based uncertainty quantification achieved 99.54% accuracy, offering probabilistic and sustainable recommendations across 22 crop classes [8]. A separate study used AI and ML models including XGBoost and PCA to improve crop yield predictions in India, resulting in higher R^2 values and reduced MSE for more effective forecasting [6]. Lastly, an investigation into agriculture's role in India's economy focused on tropical fruits like bananas and arecanut. Using ML models like XGBoost, the study achieved a high R^2 of 0.97 and lowest MAE and RMSE, supporting sustainable agriculture and policy planning [7].

2. Materials and methods

2.1. Data description and visualization

2.1.1. Data visualization

Data visualization is the process of communicating information via visual representations, transforming language and raw figures into a narrative in images that is simple for people to understand. Making data easier to access and comprehend is the main goal, since the process facilitates the finding of patterns and insights that may successfully guide decision-making.

The dataset utilized in this study is publically accessible from the Reserve Bank of India Web site <https://m.rbi.org.in/Scripts/PublicationsView.aspx?id=22492>, it encompasses 57 years of yearly agricultural statistics (1967–68 to 2023–24). This extensive and longitudinal dataset functions as a fundamental resource for developing predictive models, facilitating temporal trend analysis and time-series forecasting. Con-

sequently, it facilitates data-driven initiatives to enhance agricultural output and foster long-term sustainability within the industry.

Fig. 1 shows the year-wise production patterns of certain agricultural food grain - Groundnut, Rapeseed & Mustard, Soyabean, Coffee, Sugarcane, and Tea. This multi-line time series graph offers a whole picture of how the yield of every crop has changed over the years, exposing unique trends of increase, drop, or stability particular to every grain. Understanding the larger dynamics of agricultural output and how different crops react to different internal and external events depends on tracking these yearly changes. These comprise changing agricultural methods, climate change, regulatory actions, pressures on pests and diseases, and advances in crop management and input technology. The graph also emphasizes the relative performance of several crops throughout time, therefore enabling stakeholders to see which goods underwent notable change during specific times. Such graphic representations not only help assess previous performance but also provide empirical basis for future trend projection. It helps farms, legislators, and academics create focused plans that improve crop-specific resilience and output. Moreover, year-wise yield patterns can be linked with external events such as the adoption of improved seed varieties, changes in market demand, global commodity prices, or national-level agricultural reforms making this visualization a vital tool for both retroactive analysis and forward planning.

2.1.2. Pearson correlation heat map

Fig. 2 shows the Pearson Correlation Heat Map among six main crop yield factors, therefore revealing their linear relationships. Strong positive correlations abound, especially between Coffee and Tea ($r = 0.9413$) and Sugarcane and Rapeseed & Mustard ($r = 0.9351$), implying synchronous production patterns probably affected by common economic reasons. Groundnut shows somewhat moderate positive correlations ($r = 0.5$ – 0.6), suggesting that other or less common influences may control its production patterns. These trends are essential for machine learning applications, since elevated inter-variable correlations may signify multicollinearity, requiring feature selection procedures. On the other hand, Groundnut's quite independent actions might add special variation, hence strengthening model resilience. Especially, the heatmap shows no negative associations, suggesting a group directional shift in crop yields all throughout the dataset. This extensive dataset provides a basic resource for developing predictive models and facilitates trend analysis and time-series prediction, therefore enabling data-driven decisions to increase agricultural output and guarantee long-term sustainability.

2.1.3. The linear regression

To predict overall agricultural production from specific crop yields, Fig. 3 shows a comparative regression analysis using decision tree and linear regression models. The investigation demonstrates a consistently positive correlation among all crop types, with notably robust prediction alignment shown in Soybean, Tea, and Sugarcane, where both models nearly match the actual data points. The decision tree regressor, due to its lack of assumptions on data form and its capacity to accommodate variations across distinct segments, more effectively captures minor fluctuations and sudden shifts compared to the linear model. This is particularly evident in crops such as Groundnut and Coffee, where the correlation with total production is non-linear, resulting in the linear model exhibiting suboptimal fit and responsiveness. The comparative analysis emphasizes the fundamental constraints of linear regression in representing intricate, diverse agricultural data and showcases the enhanced flexibility of tree-based methodologies in these scenarios. The dispersed residuals and inadequate alignment in the Groundnut regression indicate its comparatively limited and uneven impact on overall yield. These findings underscore the essential need for using data-driven modeling methodologies that correspond with the structural behavior of agricultural variables to provide precise agro-economic prediction.

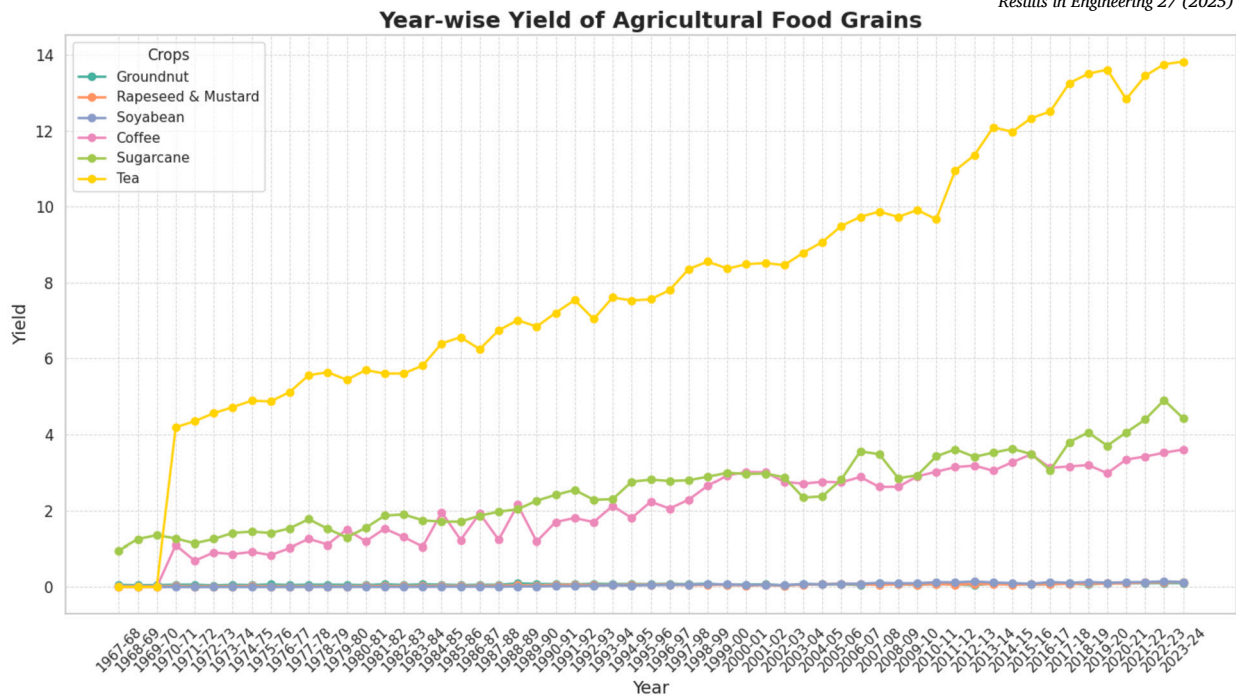


Fig. 1. Agricultural Food Grain Yield from 1967-2024.

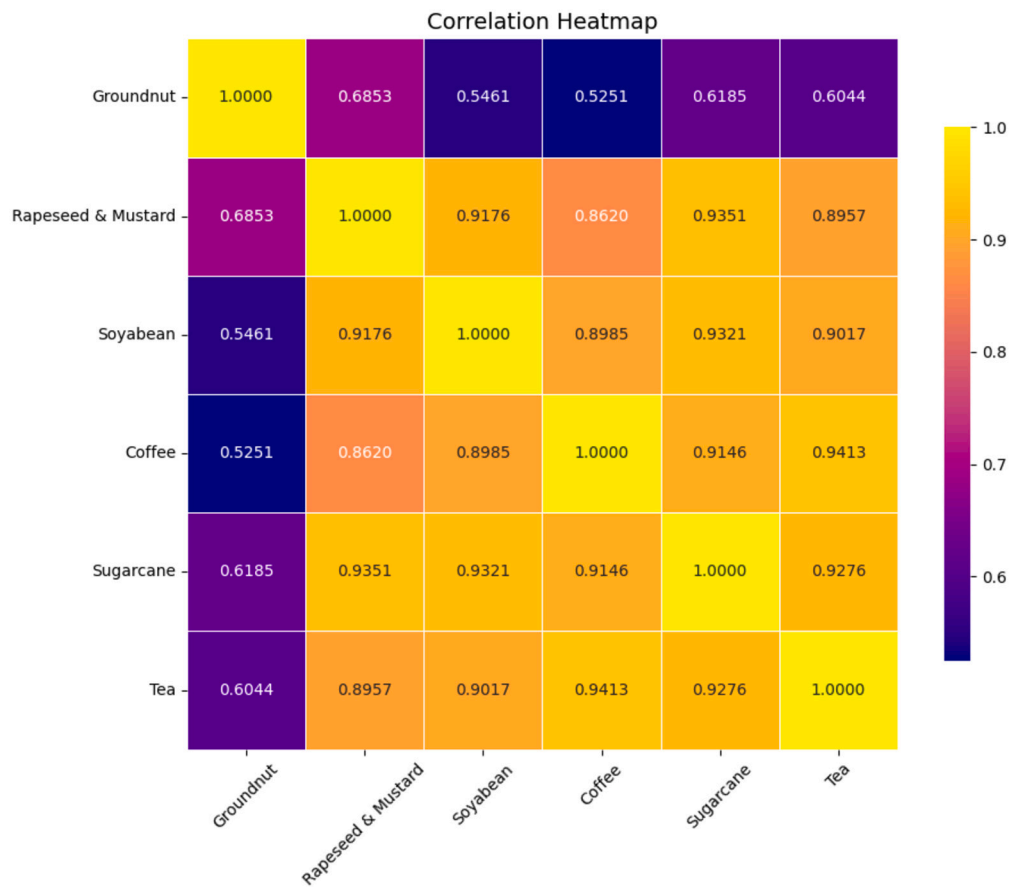


Fig. 2. Correlation Heat Map for Agricultural Food Grain Yield from 1967-2024.

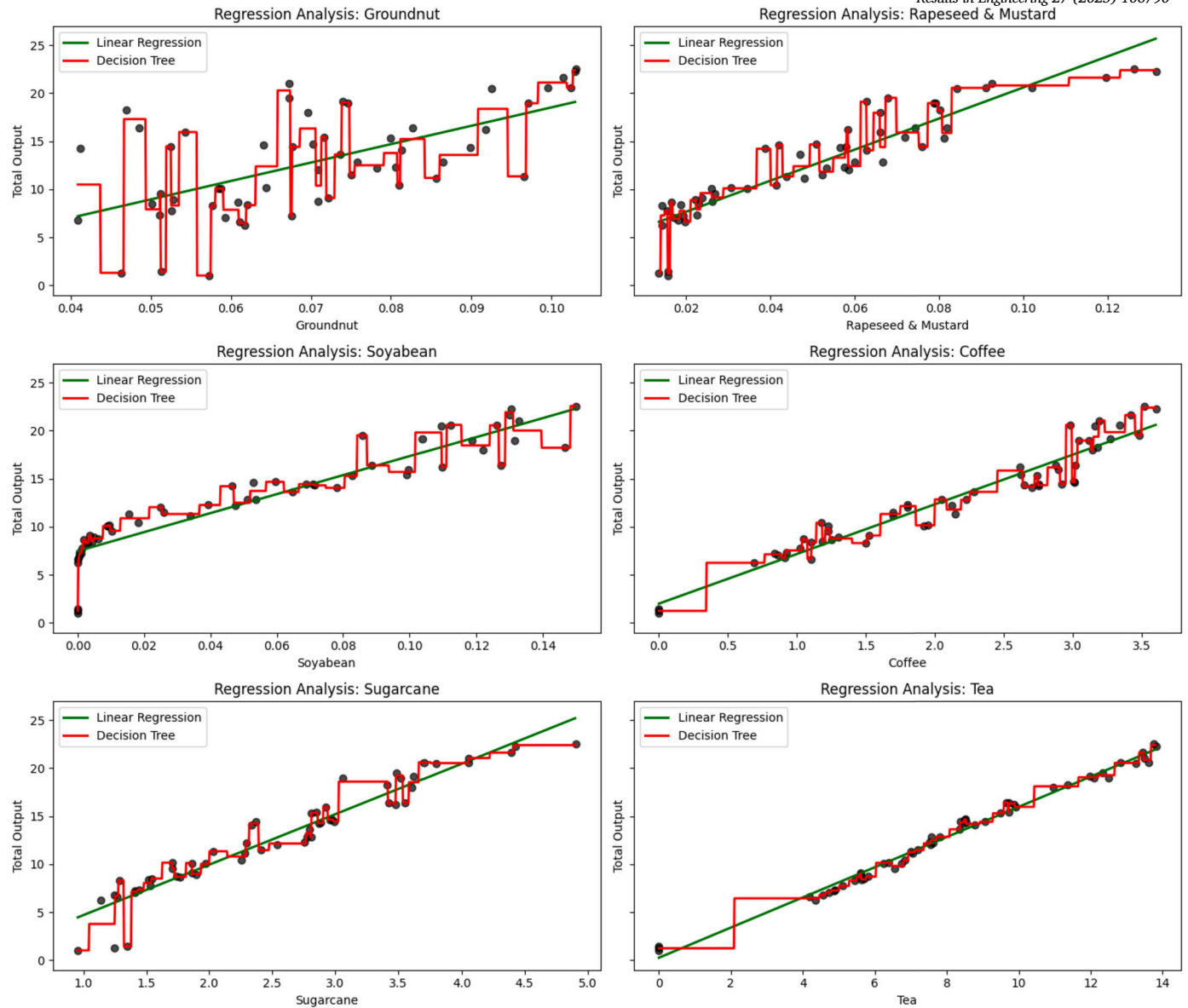


Fig. 3. Regression and Decision-Tree Plot for Agricultural Food Grain Yield from 1967-2024.

2.2. Methodology

2.2.1. Data preprocessing

The data pre-processing phase is crucial for preparing the dataset for successful machine learning analysis, providing as the foundation for obtaining accurate and interpretable results. The efficacy and dependability of machine learning algorithms are significantly dependent on the quality, consistency, and organization of the input data. This study employs historical data on agricultural food grain yield, with thorough pre-processing to rectify missing numbers, normalize distributions, and eliminate inconsistencies and outliers. These processes are crucial for improving model performance and assuring the scientific validity and robustness of the results. Data pre-processing is a fundamental step that supports the proper execution of the complete predictive modeling approach.

- **Data Cleaning:**

Finding and fixing inconsistent, inaccurate, or missing data is known as data cleansing. This procedure is essential for maintaining the dependability of machine learning results. The data cleaning procedure commences with the removal of excess spaces from col-

umn names in order to maintain consistency and prevent errors. If a 'Year' column is present, it is extracted using pattern matching and converted into a numeric format to support analysis. Missing values in numeric columns are imputed using the median, a robust measure that minimizes the impact of outliers. These steps collectively result in a clean, consistent, and complete dataset that is reliable and ready for subsequent preprocessing and model development.

- **Data Normalization:**

A basic preprocessing method used to normalize the distribution and scale of numerical variables and make sure that each feature contributes appropriately to the learning process is data normalization. In machine learning processes, especially those utilizing algorithms sensitive to feature magnitudes such as gradient-based models or distance-based classifiers-unscaled data can result in biased model behavior and diminished performance. To enhance data quality and model robustness, Z-score filtering is applied as a statistical technique for outlier detection and removal. The Z-score quantifies the number of standard deviations a data point deviates from the mean, enabling the identification of extreme values. In this process, the absolute Z-scores of all numeric features

used for outlier detection are computed, and only those rows in which all feature values lie within ± 3 standard deviations are retained.

• **Feature Selection:**

In this study, feature selection is performed by removing numerical variables that exhibit high correlation, using a threshold value greater than 0.9. This approach is intended to reduce multicollinearity, improve the interpretability of the model, and enhance its predictive accuracy. The selection process is embedded within the model training pipeline and is carried out only on the training folds during cross-validation, ensuring that the validation data remains uninfluenced. Once the optimal model parameters are determined, the same selection strategy is applied to the full training dataset to finalize the feature set. The test data is then modified to align with these selected features, maintaining consistency across datasets and ensuring the validity of the model evaluation.

• **Data Splitting:**

A key process in machine learning is data splitting, which is used to assess how well a model performs on unseen data. The dataset is often divided into training and testing subgroups to guarantee an impartial evaluation of performance. This work partitioned the agricultural dataset into an 80:20 ratio for training and testing, respectively. This method allows the model to acquire knowledge from past patterns while assessing its generalization abilities. Effective data partitioning is essential for preventing overfitting and guaranteeing the model's resilience in practical applications.

2.2.2. Machine learning technique

This study employs four distinct machine learning models to predict the total agricultural food grain yield. The comprehensive descriptions are as follows:

Gradient boost:

Gradient Boost Regression (GBR) [7,36] is a robust and extensively utilized ensemble learning method in machine learning, suitable for both regression and classification applications. It functions by constructing a robust prediction model by the iterative training of weak learners, predominantly decision trees, each designed to rectify the errors of the first. This method, termed gradient-boost trees, enables the model to refine a distinct loss function by incrementally including predictors that reduce the residuals of earlier designs. In contrast to previous boosting techniques, GBR places no limitations on the type of loss function employed, hence enhancing its flexibility.

$$G_k(x) = \sum_{i=1}^{J_k} \eta_{ik} \mathbf{1}_{R_{ik}}(x) + G_{k-1}(x) \quad (1)$$

Where

- $F_{k-1}(x)$ is the prediction from the previous (k-1) model.
- We are **updating** the model at step k by **adding corrections**.
- J_k is the number of terminal nodes (leaves) in the tree at step k .
- η_{ik} is the **optimal output value** (a constant) for region R_{ik} (leaf node i).
- $\mathbf{1}_{R_{ik}}(x)$ is an **indicator function**: it's 1 if x falls into region R_{ik} , otherwise 0.

$$\eta_{ik} = \arg \min_{\eta} \sum_{x_n \in R_{ik}} l(y_n, F_{k-1}(x_n) + \eta) \quad (2)$$

Where

- For each region (leaf node) R_{ik} , we want to find the best value η_{ik} .
- $l(\cdot)$ is the **loss function** (like squared error for regression).

- We search for the value η that **minimizes the loss** when added to the previous prediction $F_{k-1}(x_n)$ over all training samples in region R_{ik} .

Equations (1) and (2) represent the Model Update Rule and Optimal Leaf Value Calculation in Gradient Boost respectively.

XGBoost:

The XGBoost [37,38] method is a machine learning technique that is utilized for regression, learning, classification, and ranking problems. It is a very efficient and scalable approach. Using an iterative process, it constructs a collection of shallow decision trees, therefore improving accuracy by addressing flaws that were introduced by earlier models. Parallel tree building, regularization (L1 and L2) to minimize overfitting, and enhanced handling of missing and sparse data are some of the ways that XGBoost improves speed [14,34,39,40]. Its optimization reduces a regularized loss function to its minimum, striking a balance between precision and ease of use. For the purpose of resolving difficult and extensive machine learning issues, XGBoost [41,42] has gained widespread use due to its utilization of approaches such as hyperparameter tweaking and efficient processing.

The regularized objective function at iteration t is:

$$\mathcal{F}^{(t)} = \sum_{k=1}^n l(v_k, \hat{v}_k^{(t-1)} + f_t(x_k)) + \gamma(f_t) \quad (3)$$

where

- l is a differentiable convex loss function (e.g., mean squared error).
- f_t is the new tree added at iteration t .
- $\gamma(f) = \eta T + \frac{1}{2} \lambda \sum_{j=1}^N W_j^2$ is the regularization term, where:
 - N is the number of leaves in the tree,
 - W_j are the weights (scores) on each leaf,
 - η and λ are regularization parameters.

The XGBoost uses a second-order Taylor expansion to approximate the loss:

$$\mathcal{F}^{(t)} \approx \sum_{k=1}^n \left[g_k f_t(x_k) + \frac{1}{2} h_k f_t(x_k)^2 \right] + \gamma(f_t) \quad (4)$$

where

- $g_k = \partial_{\hat{v}^{(t-1)}} l(v_k, \hat{v}_k^{(t-1)})$ (first derivative, gradient),
- $h_k = \partial_{\hat{v}^{(t-1)}}^2 l(v_k, \hat{v}_k^{(t-1)})$ (second derivative, Hessian).

The best split is chosen to maximize the gain in loss reduction considering both the training loss and regularization.

LightGBM:

The process of boosting helps weak learners become stronger by integrating them into a powerful model. Any differentiable loss may be optimized using gradient-based optimization, which was first introduced by Gradient Boost Machines (GBM). Furthermore, XGBoost improved GBM by including regularization, second-order derivatives, and fast tree building. It employs level-wise tree development and is able to successfully manage sparsity, although it may have difficulty dealing with really large datasets. LightGBM [14,37,43] is a sophisticated gradient boost framework that is tailored for high-dimensional data and large-scale data. Through the utilization of Exclusive Feature Bundling (EFB) and Gradient-based One-Side Sampling (GOSS), it speeds up the training process drastically. In contrast to EFB, which effectively lowers feature space, GOSS is primarily concerned with difficult samples that contain big gradients. In order to achieve more precision, LightGBM employs a tree growth technique that is leaf-wise, in contrast to XGBoost's level-wise development. As a result of these advancements, LightGBM is now more efficient with memory, quicker, and more effective when it comes to doing complicated machine learning tasks.

Table 1
Hyperparameter settings for different Boosting Models.

Parameters / Models	Number of Estimators	Learning Rate	Maximum Depth	Subsample
Gradient Boost	100	0.05	3	0.8
XGBoost	100	0.05	3	0.8
LightGBM	200	0.05	3	0.8
CatBoost	100	0.05	3	-

CatBoost:

CatBoost [34,36], which is an abbreviation for “Categorical Boosting,” is a gradient boost technique that was created by Yandex. It is capable of handling categorical information in a natural and efficient manner without the need for human encoding. It is constructed by integrating numerous weak learners, which are often decision trees, into a strong model. Additionally, it involves the use of ordered boosting in order to minimize overfitting and guarantee stable training. CatBoost [44] is a machine learning algorithm that is very effective on both massive and small datasets. It automatically accommodates missing values, provides speedy training, and has the ability to achieve high accuracy right out of the box.

At each iteration t , CatBoost updates its model by:

$$G_t(x) = G_{t-1}(x) + \gamma \cdot g_t(x)$$

where:

- $G_t(x)$ is the updated model,
- $G_{t-1}(x)$ is the model from the previous step,
- γ is the learning rate (a small scaling factor),
- $g_t(x)$ is a new weak learner (typically a decision tree).

The residuals (or pseudo-residuals) at step t are the negative gradients of the loss function:

$$g_t^i = - \left. \frac{\partial L(v_i, G(x_i))}{\partial G(x_i)} \right|_{G(x)=G_{t-1}(x)}$$

where:

- $L(v_i, G(x_i))$ is the loss between the true label v_i and the prediction $G(x_i)$.

2.2.3. Model implementation

Hyperparameter tuning

The process of determining the optimal combination of parameters to maximize the performance of a model is referred to as hyperparameter tuning. To differentiate themselves from the internal model parameters, which are learnt, hyperparameters are values that are set individually when learning starts. The process of tuning is essential since it helps to increase accuracy, boost model generalization, and strike a balance between underfitting and overfitting problems. In this study, hyperparameter tuning is carried out using GridSearchCV [17,27,28,45] in conjunction with 5-Fold [15,32,35] cross-validation [24,29–31,45,46]. This approach systematically explores a predefined set of hyperparameter combinations while ensuring robust performance evaluation across multiple data splits.

The Hyperparameter setting for different Boosting models are displayed in Table 1

This study employs a consistent grid of hyperparameters across four boost models - Gradient Boost, XGBoost, LightGBM, and CatBoost - to ensure fair and uniform comparisons. Each model is set to use 100 trees, striking a balance between training efficiency and model complexity. The learning rate, set at 0.05, controls the extent to which each new tree corrects the errors of its predecessors, enabling the model to learn more gradually and with greater stability. A tree depth of 3 is chosen to keep the models relatively simple and to support better generalization. Furthermore, a subsample ratio of 0.8 is applied, meaning that

each tree is trained on a randomly chosen 80% subset of the training data. This introduces an element of randomness, which enhances model robustness and helps prevent overfitting. These hyperparameter values are likely chosen to ensure a balanced trade-off between model accuracy and training time.

Model training:

This implementation presents a comprehensive and efficient machine learning pipeline is shown in Fig. 4.

This pipeline combining preprocessing, model tuning, interpretability, and SHAP Analysis. The process begins with the imputation of missing values and outlier removal using the z-score method, followed by the elimination of highly correlated features through a custom transformer designed to improve model generalization. These steps are embedded within a scikit-learn pipeline that includes feature scaling and Regressor models. Hyperparameter optimization is performed via GridSearchCV, which systematically evaluates combinations of key model parameters such as learning rate, number of estimators, maximum tree depth, and subsampling ratio by training and validating the model multiple times using 5-fold cross-validation. In this process, the training dataset is split into five equal parts (folds), where the model is trained on four folds and validated on the remaining one, repeating this process five times so that each fold serves as a validation set once. The average performance across all folds is used to select the best parameter combination, ensuring reliable generalization and avoiding overfitting to any single subset of data. The best model is retrained using the optimal parameters and incorporates early stopping based on validation performance to mitigate overfitting. Model evaluation is carried out using multiple metrics, including MSE, MAE, RMSE, adjusted R^2 , and bias indicators, across both training and testing sets. Visual comparisons between actual and predicted values are generated using scatter and line plots to provide a tangible view of model performance. Finally, SHAP-based explainability techniques are applied to quantify and visualize feature contributions through summary, bar, force, and dependence plots, offering deep insights into model behavior and enhancing interpretability.

3. Comparative results and discussion

3.1. Evaluating the performance of the models

Error metrics are extremely important in the field of machine learning because they offer quantifiable measurements of the disparity between the outcomes that were anticipated and those that actually occurred. They create a framework that is objective for evaluating the quality of models, identifying systemic biases, and diagnosing problems such as underfitting and overfitting. Additionally, error metrics make it easier to choose, evaluate, and optimize models by methodically pointing out each one's advantages and disadvantages. In addition to this, they make it possible for practitioners to monitor the performance of the model during the training process and to evaluate its resilience on data that has not been explored before. In the end, the use of error measures in a consistent manner guarantees that machine learning models are correct in their predictions. Table 2 shows the error functions used in this study.

Where:

- m = number of observations
- v_k = actual (true) value for the k -th observation

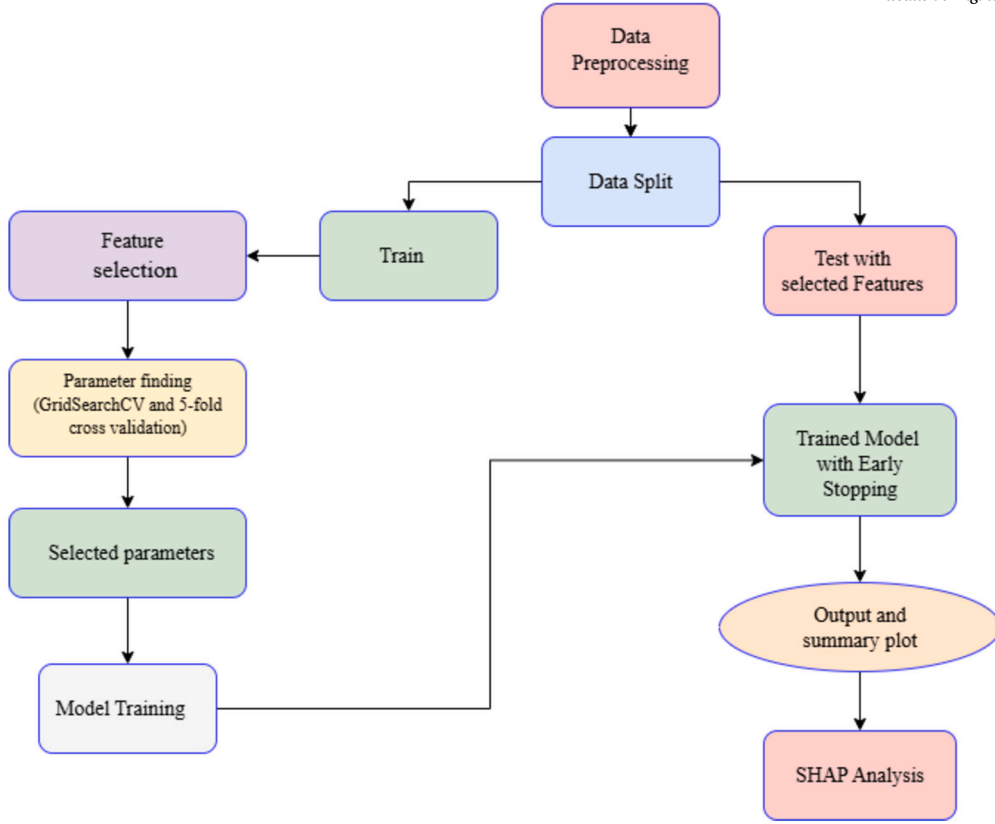


Fig. 4. Over all Pipe Line of the Model.

Table 2
Error Metrics.

Metric	Formula	Reference
Mean Error (ME)	$ME = \frac{1}{m} \sum_{k=1}^m (v_k - \hat{v}_k)$	[7,14,33]
Mean Squared Error (MSE)	$MSE = \frac{1}{m} \sum_{k=1}^m (v_k - \hat{v}_k)^2$	[7,14,33,34,47,48]
Root Mean Squared Error (RMSE)	$RMSE = \sqrt{\frac{1}{m} \sum_{k=1}^m (v_k - \hat{v}_k)^2}$	[7,14,33,34,39]
Mean Absolute Error (MAE)	$MAE = \frac{1}{m} \sum_{k=1}^m v_k - \hat{v}_k $	[7,14,33,34,47,48]
R-Square (R^2)	$R^2 = 1 - \frac{\sum_{k=1}^m (v_k - \hat{v}_k)^2}{\sum_{k=1}^m (v_k - \bar{v})^2}$	[7,10,11,14,33,34,49]
Mean Percentage Error (MPE)	$MPE = \frac{1}{m} \sum_{k=1}^m \left(\frac{v_k - \hat{v}_k}{v_k} \times 100 \right)$	[7,14,33,34]

- $\hat{v}_k$ = predicted value for the k -th observation
- $(v_k - \hat{v}_k)$ = individual prediction error (residual)
- $|v_k - \hat{v}_k|$ = absolute value of the prediction error
- $\bar{v}$ = mean of the actual values

3.2. Comparative evaluation of the models

In agricultural-based economies like India, higher food grain yield contributes to general economic growth by raising the incomes of farmers, eliminating poverty, and promoting rural development. This is especially true in these countries. The primary objectives of this research are to improve predictions of total agricultural food grain yield output and extract valuable information from a sizable dataset. Publicly available on the Reserve Bank of India website, the dataset used in this research includes 57 years of annual agricultural statistics (1967–1968 to 2023–2024). Groundnut, Rapeseed and Mustard, Soybean, Coffee, Cotton, Raw Jute and Mesta, Sugarcane, and Tea are among the vital

Table 3
Results for Different ensemble Models during Testing.

Models	ME	MSE	RMSE	MAE	R^2	MPE
Gradient Boost	0.4945	2.1830	1.4775	0.9142	0.9550	0.0093
XGBoost	0.8769	3.6540	1.9115	1.1228	0.9247	0.0401
LightGBM	1.0512	10.2556	3.2024	2.5054	0.7885	0.6203
CATBoost	0.6039	3.9699	1.9925	1.5439	0.9181	0.1979

crops whose yields are thoroughly documented in billion metric tones. By assessing many predictive models using a range of metrics, comprehensive insights into their training and testing prediction potential are provided. The models' comparison in terms of error metrics during testing is shown in Table 3.

A visual comparison of four popular boosting models - Gradient Boost, CatBoost, XGBoost, and LightGBM is shown in Fig. 5. Plotting predicted and actual values for each model during training and testing

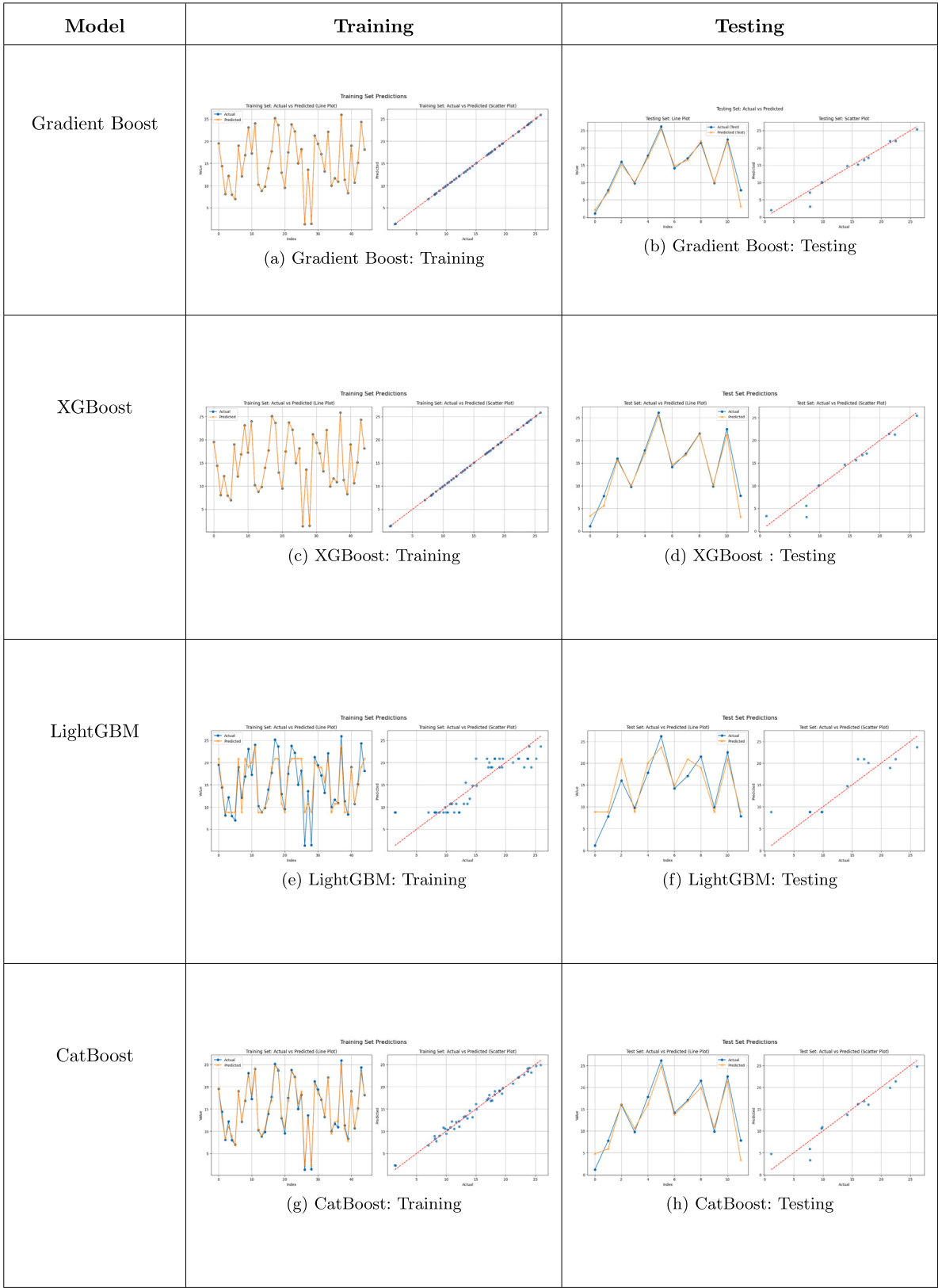


Fig. 5. Actual vs Prediction plots for Training and Testing across Boosting Models.

enables an easy evaluation of the model's performance and capacity for generalization. The testing plots show how successfully the learnt patterns are applied to new data, while the training plots assist confirm how well the models have learned from the data. Higher accuracy and less bias are suggested by a model whose predictions are closely aligned along the diagonal, or ideal line. This figure makes it possible to quickly compare learning behavior, overfitting tendencies, and prediction consistency across several boosting techniques by arranging the models in a tabular structure.

Evaluation during training

Gradient Boost is an extremely successful batch learning approach that constructs decision trees in an ordered manner, enabling each new tree to fix the mistakes of the ones that came before it. With an R^2 value of 1.0 and all error metrics ME, MSE, RMSE, MAE, and MPE equal to zero, it produced flawless outcomes throughout training. The model exhibits a remarkable capacity to identify the fundamental patterns present in the data. The model's great learning capacity and ability to provide accurate predictions when appropriately verified are reflected in this level of accuracy. The training dataset's actual and predicted values are highlighted in Fig. 5. a. The scatter plot's blue dots form a straight line along a red dashed diagonal, highlighting how closely forecasts match real data points, while the blue line denotes actual and orange for prediction.

XGBoost is a scalable and highly efficient gradient boost system that is well-known for its ability to handle missing data, perform regularization, and allow parallel processing. With an R^2 value of 1.0 and nearly zero error metrics (ME, MSE, and MAE) on the training data, it demonstrates exceptionally high prediction accuracy. This performance level indicates that the model has learnt the training data rather effectively, much as Gradient Boost. In Fig. 5. c shows the model obtains an almost perfect match between projected and actual values on the training set, with minimal variation from the red reference line, as seen by the line plot and scatter plot.

Microsoft created LightGBM, a quick and memory-efficient gradient boosting method that works especially well with big datasets. The training metrics display a mean absolute error (MAE) of 2.1024, a mean squared error (MSE) of 7.7001, a root mean squared error (RMSE) of 2.7749, a mean percentage error (MPE) of 0.2189, and an R^2 of 0.7921. The model does not fully capture the training data, as shown by the R^2 value being less than 1, and the performance on the training data is moderate with discernible mistakes. This model's performance is shown in Fig. 5. e, where the left subplot demonstrates a strong match between the predicted and actual values, with only minor variations. This is further supported by the right subplot, which shows that the majority of the points are grouped close to the red diagonal line, indicating a high degree of agreement between the actual and expected values with very slight prediction errors.

The purpose of the CatBoost is to minimize overfitting, handle categorical variables well, and require less preprocessing of the data. Mean Squared Error (MSE) of 0.4373, Root Mean Squared Error (RMSE) of 0.6613, Mean Absolute Error (MAE) of 0.5369, R^2 of 0.9882, and Mean Percentage Error (MPE) of 0.0317 are the metrics recorded during training. With minimal errors and a high R^2 , the model performs well, showing that it isn't perfect but still quite excellent. The line plot (left) in Fig. 5. g illustrates how projected values follow real ones, while the scatter plot (right) shows a close clustering of points around the red reference line, confirming the model's good performance.

Comparison on testing period

During the testing phase, the Gradient Boost model exhibits exceptional predictive accuracy and reliability, with low error values, such as an MSE of 2.1830 and an RMSE of 1.4775, suggesting accurate and consistent predictions, and negligible overprediction bias (ME = 0.4945). While the R^2 value of 0.9550 indicates that the model explains 95.5% of the variation in the target variable, indicating good explanatory power, the MAE of 0.9142 verifies consistently low absolute errors across the

training data. Furthermore, the MPE of 0.0093 indicates superior scaled accuracy with an average relative error of less than 1%.

The prediction and actual results are displayed in both line and scatter graphs in Fig. 5. b, which emphasizes how well the expected and actual values match. While the scatter graph provides a more in-depth look at individual data points, highlighting the model's skill in identifying underlying patterns and reducing prediction errors, the line graph shows the general trend and consistency of the predictions.

The XGBoost model performs well, showing a slight overprediction bias with a Mean Error (ME) of 0.8769. Although they fall within acceptable bounds, its Mean Squared Error (MSE) of 3.6540 and Root Mean Squared Error (RMSE) of 1.9115 indicate greater deviations than Gradient Boost. With forecasts varying by an average of little over one unit, the Mean Absolute Error (MAE) of 1.1228 indicates respectable accuracy. Strong predictive power is confirmed by the model's ability to explain 92.5% of the variation in the target variable, as indicated by its R^2 value of 0.9247. Although not the lowest of the models, the average relative error of 4% is shown by the Mean Percentage Error (MPE) of 0.0401, which is satisfactory. The line plot (left) in Fig. 5. d indicates a low difference between the actual and predicted values, and the scatter plot (right) supports this finding by showing the majority of data points close to the ideal red line, suggesting that the test set performance is high.

The performance of the LightGBM model is the lowest of all the models that were assessed. It routinely predicts values more than one unit above the real, exhibiting the largest overprediction bias with a Mean Error (ME) of 1.0512. It has the greatest Root Mean Squared Error (RMSE) of 3.2024 and Mean Squared Error (MSE) of 10.2556, which suggests poor generalization and frequent huge deviations. The mean absolute error (MAE) of 2.5054, which shows a large deviation between forecasts and actual data, further supports the low accuracy. Only 78.9% of the variation in the target variable is captured by an R^2 value of 0.7885, which indicates the lowest explanatory power. Finally, with predictions that are on average 62% off from genuine values, the Mean Percentage Error (MPE) of 0.6203 indicates a very low scaled accuracy. Fig. 5. f illustrates how effectively the model depicts the general trend of the test data, with overlapping curves visible in the line plot. The scatter plot shows that there are a few outliers, which indicate a little prediction variation, even if many points are located around the red reference line.

In comparison to other models, the CatBoost model performs minimally. Its moderate overprediction tendency is indicated by its Mean Error (ME) of 0.6039. While still superior than LightGBM, the Mean Squared Error (MSE) of 3.9699 is greater than both XGBoost and Gradient Boost. Its Mean Absolute Error (MAE) of 1.5439 indicates a wider range of prediction oversights, while its Root Mean Squared Error (RMSE) of 1.9925 indicates that forecasts often differ by over 2 units. Although it is somewhat less than XGBoost, the model's R^2 score of 0.9181 suggests that it can explain 91.8% of the variation in the target variable. However, compared to the best-performing models, the Mean Percentage Error (MPE) of 0.1979, or over 20%, indicates significantly lower scaled accuracy. Visual proof of the model's excellent performance is shown in Fig. 5. h. The scatter plot, which shows data points around the red $y = x$ line, supports the line plot's indication of tight synchronization between actual and anticipated values.

Shapley additive exPlanations (SHAP) analysis:

A post hoc explainability method based on cooperative game theory is called SHAP [17,50]. It is very helpful for analyzing machine learning models since it determines the value of each feature by calculating its contribution to the final prediction. By allocating the model output across all input characteristics according to their marginal contributions, SHAP values are calculated. SHAP's [13,14,51] capacity to describe both local and global feature consequences is one of its special advantages. While local SHAP values describe the influence of input variables for particular data points, global SHAP values measure the overall impact of each feature on the model's predictions.

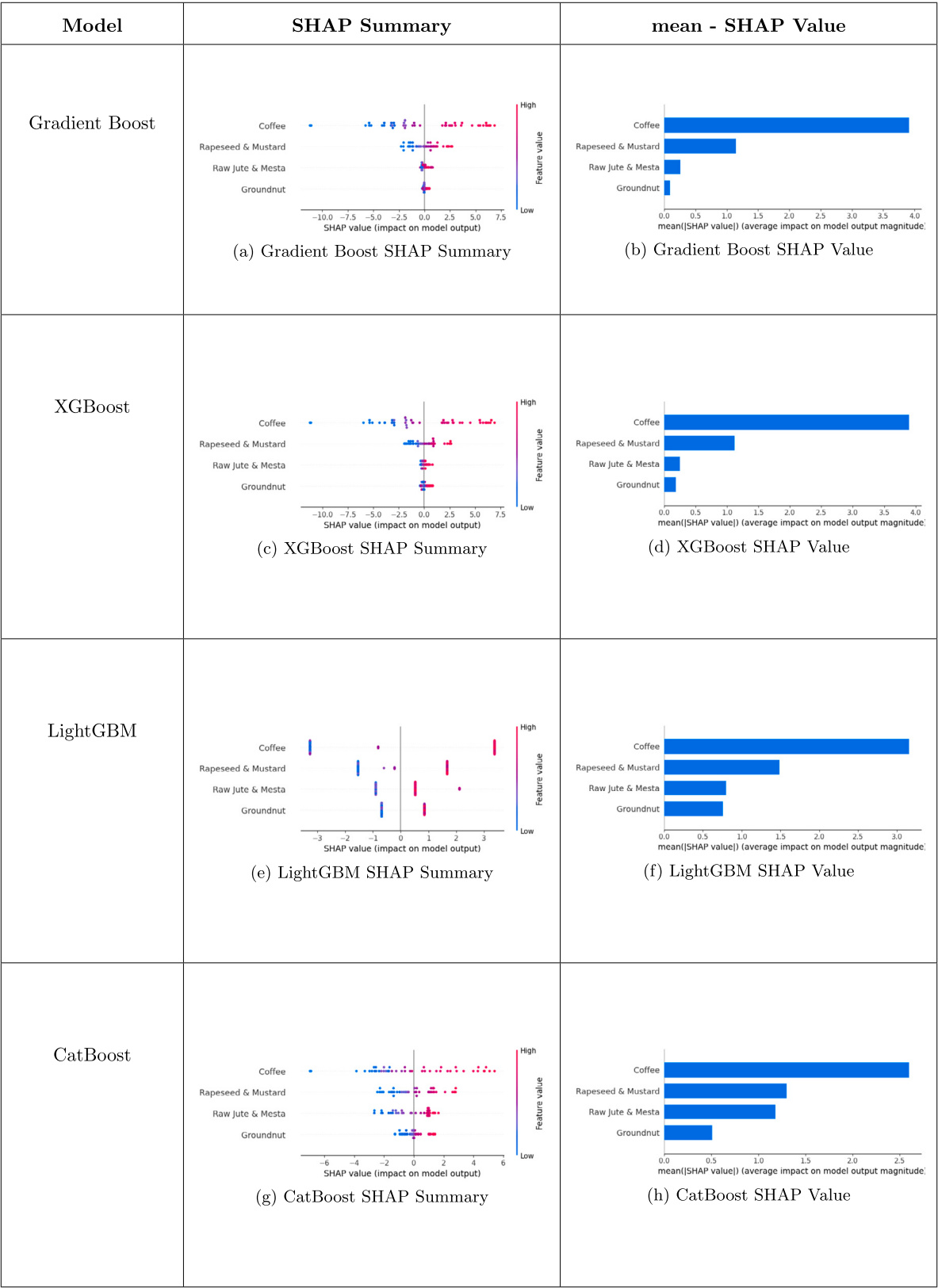


Fig. 6. SAHP Value and Summary - across Boosting Models.

The SHAP beeswarm summary plots (Figs. 6. a, c, e, and g) for each model visualize the distribution of SHAP values across all features, providing insights into how individual feature values influence predictions. In a SHAP beeswarm plot, the color of each dot represents the value of the feature for a specific data point, where pink indicates a high feature value and blue indicates a low feature value. In the Gradient Boost model (Fig. 6.a), high values of “Coffee” (in pink) strongly associate with positive SHAP values, indicating a consistent and clear positive impact on the model’s output, while low values contribute negatively. Other features, such as “Rapeseed & Mustard” and “Groundnut,” display weaker and more scattered SHAP distributions. In the XGBoost model (Fig. 6.c), SHAP values for “Coffee” show both strongly positive and negative impacts, reflecting a more complex, non-linear, and context-dependent influence likely due to captured interactions or conditional effects. “Rapeseed & Mustard” also exhibits notable variability. The CatBoost model (Fig. 6.e) again shows that high “Coffee” values contribute positively to predictions, while lower values result in negative SHAP contributions. In this model, “Rapeseed & Mustard” and “Raw Jute & Mesta” present more symmetric and centered SHAP distributions, suggesting that their influence depends on value range and context, indicating a nuanced relationship. Lastly, the LightGBM model (Fig. 6.g) reveals a clear and consistent positive trend for “Coffee” higher values lead to higher SHAP contributions, while lower values reduce the prediction output. This consistency highlights LightGBM’s strength in capturing monotonic feature relationships, where features like “Coffee” exert a predictable directional impact on model output.

The mean - SHAP value plots clearly illustrate the results across all four methods Gradient Boost (Fig. 6.b), XGBoost (Fig. 6.d), CatBoost (Fig. 6.h), and LightGBM (Fig. 6.f). In the Gradient Boost model, “Coffee” shows the highest mean SHAP value, followed by “Rapeseed & Mustard,” while “Raw Jute & Mesta” and “Groundnut” contribute minimally, indicating limited influence on model predictions. Similarly, in the XGBoost model, “Coffee” again leads in mean SHAP impact, with “Rapeseed & Mustard” ranking second, while the other two crops remain relatively insignificant. In the CatBoost model, “Coffee” maintains its dominant role, but the contributions of “Rapeseed & Mustard,” “Raw Jute & Mesta,” and “Groundnut” appear more balanced. The LightGBM model also ranks “Coffee” as the most important variable, though with a slightly lower mean SHAP value. “Rapeseed & Mustard” follows, while “Raw Jute & Mesta” and “Groundnut” exert moderate influence. LightGBM captures more subtle feature effects, enhancing the interpretability of crop contributions.

4. Conclusion

This study evaluates the performance of four advanced machine learning models Gradient Boost, XGBoost, CatBoost, and LightGBM - for predicting total agricultural food grain yield. Each model was trained and validated using a consistent pipeline involving GridSearchCV, 5-fold cross-validation, early stopping, and SHAP-based interpretability. Based on the experimental results, **Gradient Boost** demonstrated the highest prediction accuracy, achieving the lowest error values on the test set (ME = 0.4945, MSE = 2.1830, RMSE = 1.4775, MAE = 0.9142, MPE = 0.0093, R^2 = 0.9550), followed by **CatBoost** (R^2 = 0.9181, RMSE = 1.9925) and **XGBoost** (R^2 = 0.9247, RMSE = 1.9115). **LightGBM** showed comparatively lower performance (R^2 = 0.7885, RMSE = 3.2024), suggesting its limitations with this dataset.

Based on these results, Gradient Boost stands out as a robust and reliable model for food grain yield forecasting, with strong generalization ability and minimal prediction errors. The integration of SHAP analysis revealed that crops like Coffee and Rapeseed & Mustard consistently contribute the most to yield predictions, enabling greater transparency and trust in model outcomes.

These findings underscore the practical utility of ensemble machine learning models in improving the accuracy and reliability of agricultural yield prediction. For stakeholders such as farmers, policymakers,

and agribusiness planners, such predictive insights can directly inform crop selection, planting schedules, input allocation, and long-term food security planning. The model’s interpretability through SHAP analysis enables a better understanding of key crop contributions, supporting transparent and evidence-based decision-making. By aligning model outcomes with actionable agricultural strategies, this study offers a valuable tool that can enhance data-driven interventions and guide policy toward sustainable and resilient food production systems.

CRedit authorship contribution statement

Nirmaladevi S: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jagatheswari S:** Supervision.

Ethics approval

Not applicable. This study did not involve human participants or animals.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and material

The dataset used and analyzed during the current study is publicly available Reserve Bank of India data. The link is provided in section 2.1.1.

Code availability

The code used for model training and evaluation is available from the corresponding author upon reasonable request.

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Declaration of competing interest

The authors declare no conflicts of interest or competing financial interests related to the research or publication of this paper.

Data availability

Data will be made available on request.

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